

# “Efficiency In **SOLAR ENERGY** Is More About **INVESTMENT RATIONALE** Than Resource Limitation”

Indian utility-scale solar company SolarArise claims to offer investors long-term, cash-generating assets and anticipates advancements in energy storage technologies to enable 24-hour clean power at competitive prices. In a freewheeling chat with EFY's Yashasvini Razdan, SolarArise's Co-founder Tanya Singhal divulged interesting details about the company's business model, challenges, technological innovations, and future plans of contributing to the country's renewable energy goals



**TANYA SINGHAL**  
CO-FOUNDER, SOLARARISE

## **What does SolarArise do?**

SolarArise is focused on building large solar plants to generate and supply solar power directly to the national grid. It secures land with ample sunshine, procures solar modules, raises funds for equity and debt, constructs the plant, builds lines for power evacuation to the nearest substation, and operates the plant for 25 years with an intent to reduce the cost of solar power generation through improved technology over the years.

## **What business opportunity does SolarArise present to investors?**

SolarArise offers investors a chance to own long-term, cash-generating assets. Approximately 95% of the investment is made upfront in the capital-intensive solar business, with minimal annual operational and maintenance costs. The generated revenue significantly contributes to a reliable cash flow, making it appealing to investors such as pension funds or life insurance corporations. These inves-

tors, seeking annuity income with a long-term horizon, value the consistent, double-digit annual returns that solar investments can provide over 10, 15, or even 25 years.

## **Solar panels are said to have less efficiency. How does that compare with traditional energy generation sources?**

The concept of efficiency in the context of solar energy is fundamentally different from traditional energy generation sources. In conventional sources, efficiency matters because you deal with finite resources burned or extracted from the earth. With solar energy, the sun is always there; if you don't harness it, you simply miss out on its potential. The real question of efficiency in solar energy depends on the returns on the investment made in solar infrastructure. Are we generating the expected amount of electricity annually for the capital invested? It is about maximising the power generated from an

existing, ever-present source. The more we invest in solar energy, the less we rely on inefficient, polluting sources like coal. So, in solar energy, efficiency is more about investment rationale than resource limitation.

## **What is scheduling, and why is it necessary?**

Government mandates compel solar producers to schedule power output for the next 24 hours in one-hour slots with predictions every 15 minutes, balancing energy supply in the grid. A permissible deviation threshold exists, and significant underproduction or overproduction may lead to penalties. This practice, applicable to solar and wind energy, is called forecasting and scheduling.

## **Are IoT devices or sensors used in your solar energy projects?**

Sensors in solar plants differ from those in smart grids, being plant-specific and focusing on individual solar module performance. Each module